

FINAL YEAR PROJECT

PARKFLOW POS: INTEGRATED POINT OF SALE AND BACKOFFICE SYSTEM FOR PARKING OPERATIONS

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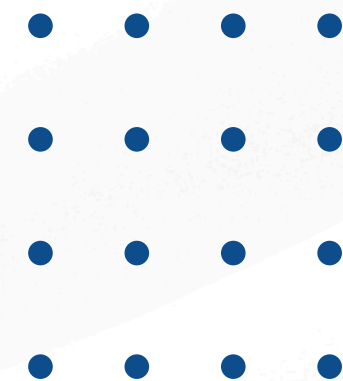




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01 Introduction

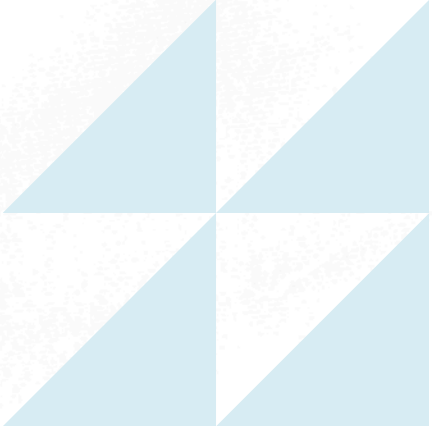
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Introduction

Host Organization



Based in Mégrine, specializing in parking management



Products: Park & Charge, Taxi Management Platform



Clients: airports, universities



Upcoming deployment: Cairo International Airport

Project Context

Core Elements



Existing System and Limitations

- Slow at processing
- Lack of Plate processing
- No Image handling
- Online mode only

10.06.2026 - 22:55 Shift no. 211 - super user

booked articles:

Interim result:

Still due to pay:

Change money:

Please enter article no.:

  Wind

702 - ON-LINE -



Proposed Solution

01 POS System



- Real-time vehicle payment processing
- Shift management, receipt printing, ...
- Offline mode
- Licence plate as media

02 Backoffice Platform



- Dynamic configuration
- Management of cashiers, articles, barriers, cameras, ...
- Full audit trail and monitoring

Implementation

System Actors



Cashier

Operates the POS System

☒ human actor



Administrator

Manage Configuration

☒ human actor



Camera service

LPN detection

☐ Non-human



Barrier service

Gate control

☐ Non-human



Printer service

Receipt printing

☐ Non-human



Tariff service

Fee calculation

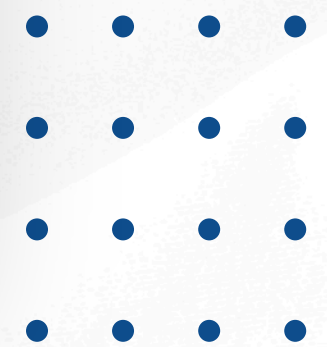
☐ Non-human



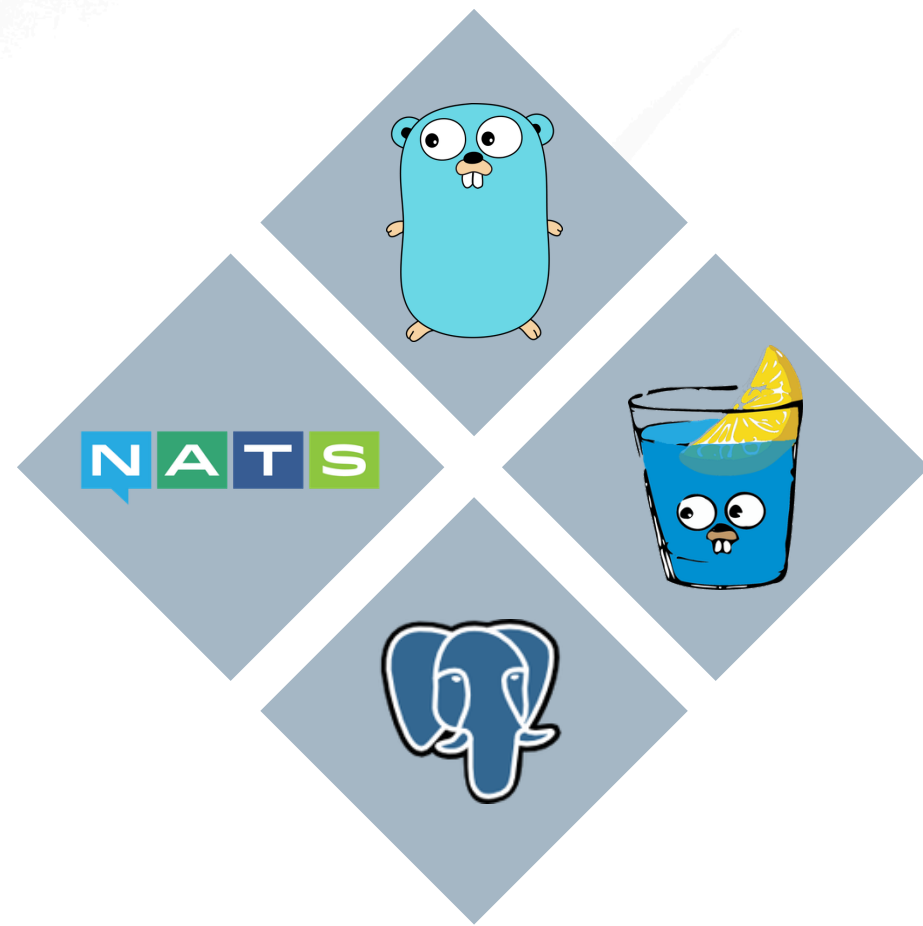
PMS service

Parking management

☐ Non-human



Technical Choices



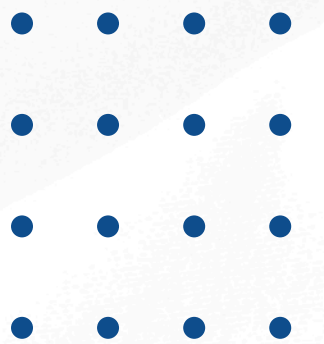
Backend Technologies



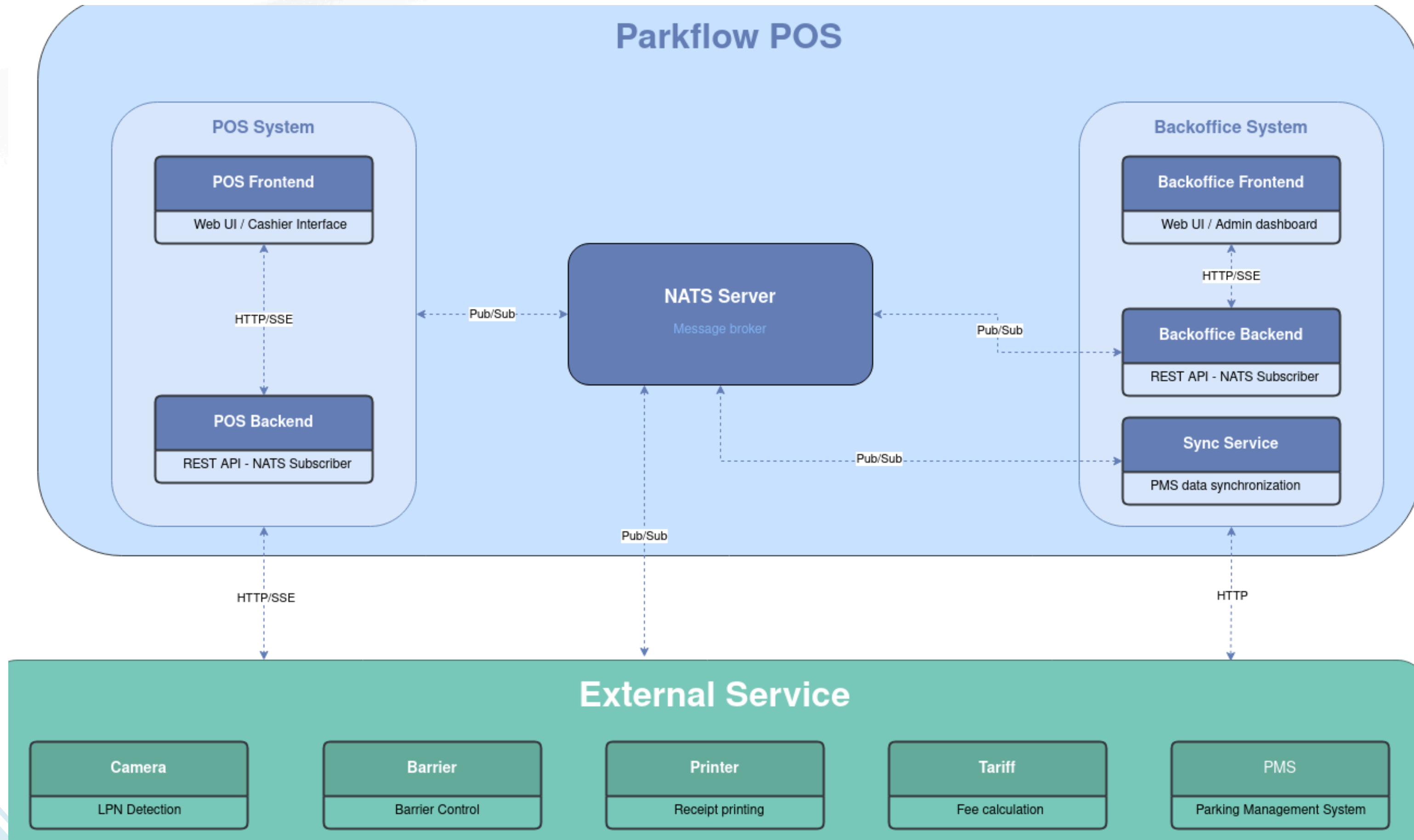
Frontend Technologies



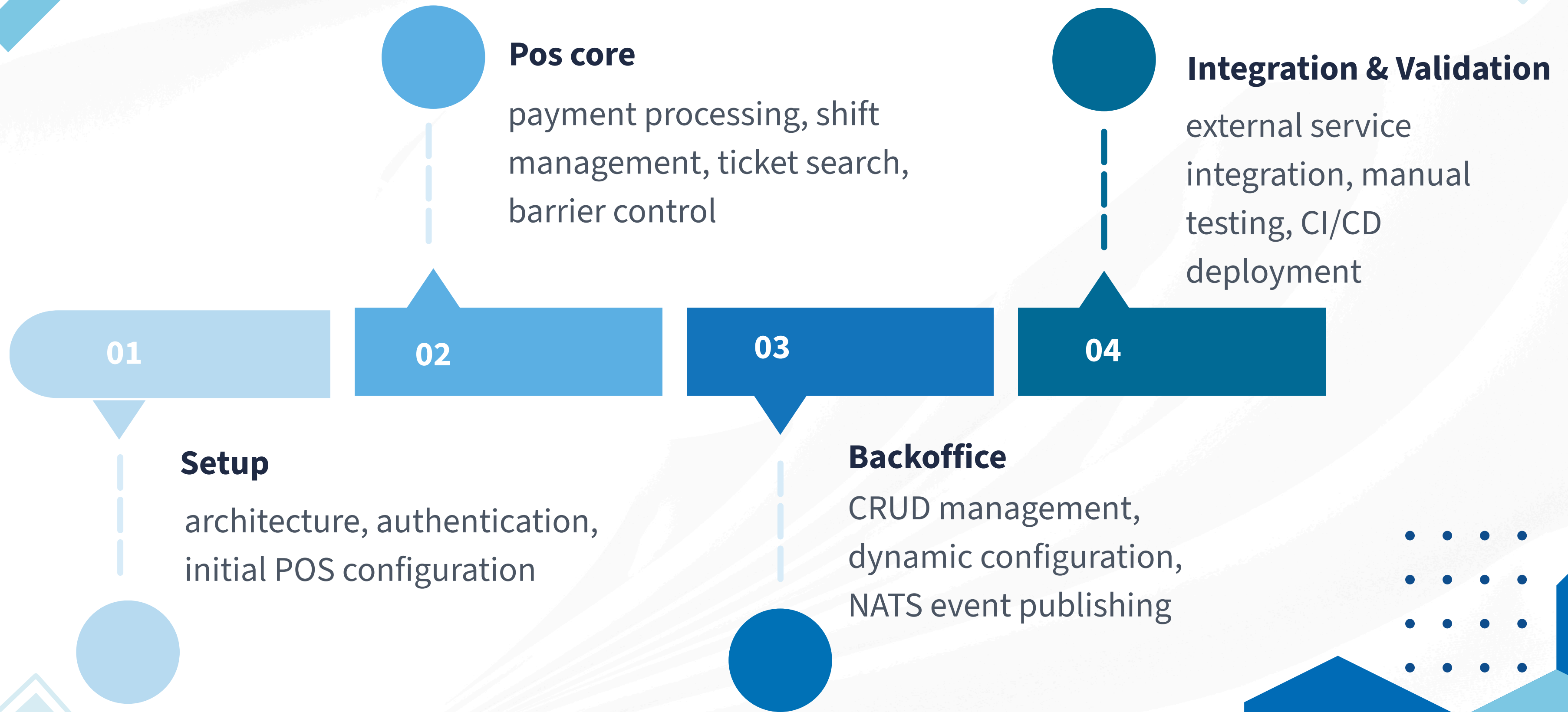
**Development Tools &
Infrastructure**



System Architecture



Development Approach



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Demo

Demo

→ Demo1: Setting up new POS terminal

The screenshot displays the 'Posify Backoffice' interface with the 'Pos Management' section active. A modal window titled 'Add Pos' is open, showing the 'General' tab. The form includes the following fields and options:

- carpark ***: A dropdown menu with 'Parking Central 1' selected.
- Pos Name ***: A text input field with the placeholder 'Enter the POS name'.
- PRODUCTS**: A section with a 'Short Term Parker' tag and a 'Select products' dropdown.
- POS OPTIONS**: A section with three toggleable options: 'Standalone' (disabled), 'Payment method' (set to 'All'), and 'With barrier' (disabled).

At the bottom of the modal are 'Cancel' and 'Submit' buttons. The background interface shows a sidebar with navigation links (Dashboard, Roles, Users, Cashiers, Carparks, Pos Management, Cameras, Articles, barriers, Barrier Control, Reporting, Audits, Settings) and a main content area with a 'Total Points' summary (70) and a table of POS terminals. The table has columns for 'Carpark ID', 'Status', 'Created At', and 'Actions'. The user 'Admin Admin' is logged in.

Interface Preview

→ Demo 2: Screenshot Exit car, payment process scenario

Posify

Payment

Shift Report

SYSTEM ONLINE

English

Pause Shift

Close Shift



LICENSE PLATE

AFH147

EPAN

LPN_8f3d7f1e-4b91-4f62-a0f0-6a1e7d5d8c4b

CARD TYPE

Short Term Parker

ENTRY TIME

2026-06-09 06:05:32

PARKING DURATION

8h 12m

LICENSE PLATE

AAA-0000

FROM

01/01/2026 00:00:00

TO

09/06/2026 23:59:59

SEARCH

ARTICLES / VALIDATION

F1



STAFF TICKET

ACCESS GRANTED FOR AUTHORIZED STAFF VEHICLES

F3



LOST TICKET

CHARGE APPLIED WHEN CUSTOMER LOSES PARKING TICKET

F6



FIRST 30 MINUTES FREE

COMPLIMENTARY PARKING FOR THE FIRST 30 MINUTES OF STAY

F2



POLICE FINE

FINE ISSUED DUE TO VIOLATION REPORTED BY POLICE

F4



10% DISCOUNT

STANDARD 10 PERCENT DISCOUNT APPLIED TO TOTAL

F7



DAMAGED TICKET

TICKET UNREADABLE OR PHYSICALLY DAMAGED

BILLING

CLEAR ALL



SHORT_TERM_PARKER

2.2 EGP

X



FIRST_30MIN_FREE

-0.44 EGP

X

SUBTOTAL

1.76 EGP

DISCOUNTS

0.44 EGP

TAX (20%)

0.293 EGP

TOTAL

1.76 EGP



BARRIER



ERROR



AUTO PRINT

PAYMENT METHOD



CASH



CARD

POS 7 | POS_001 | AK Ayoub Krimi | CONNECTED

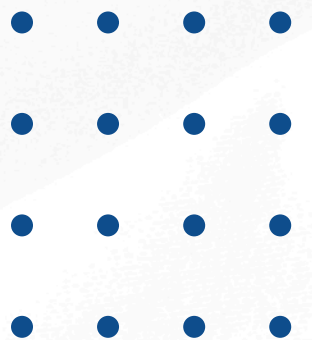
CONFIGURATIONS

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Conclusion and Perspectives

Conclusion

- ✓ Fully functional POS system
- ✓ Complete backoffice platform
- ✓ Event-driven NATS architecture
- ✓ PMS-agnostic design
- ✓ Validated through manual testing
- ✓ Ready for production deployment



Perspectives

1

Automated Testing



2

Advanced Analytics Dashboard



3

Additional Operational Scenarios



THANK YOU

Presented by : Ayoub Krimi

